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- A 3D coordinate system with axes x , y , and z . The vertices of the tetrahedron are labeled: $A(4,0,0)$, $B(4,4,0)$, $C(0,4,0)$, and $D(2,2,4)$. The base is a square in the xy -plane. A point M is marked on the edge BC . Dashed lines represent hidden edges and the projection of D onto the base.



Application

- 5 In a circus, a wire, A , is attached to the top of a 30 ft pole. Wire B is used for performers to walk up to the tight wire, 10 ft above the ground. The wires are attached 40 ft from the pole. Find the angle θ between the wires. Round your answer to two decimal places.

